Integrating using inverse trig

Basic skills

Q1. a) $y = \arctan x \Rightarrow x = \tan y$

$\Rightarrow \dfrac{dx}{dy} = \sec^2 y = 1 + \tan^2 y = 1 + x^2$

$1 + \tan^2 y = \sec^2 y$

so $\dfrac{dy}{dx} = \dfrac{1}{1+x^2}$

b) $y = \arcsin x \Rightarrow x = \sin y$

$\Rightarrow \dfrac{dx}{dy} = \cos y = \sqrt{1-\sin^2 y} = \sqrt{1-x^2}$

so $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-x^2}}$

c) $y = \arccos x \Rightarrow x = \cos y$

$\dfrac{dx}{dy} = -\sin y = -\sqrt{1-\cos^2 y} = -\sqrt{1-x^2}$

so $\dfrac{dy}{dx} = \dfrac{-1}{\sqrt{1-x^2}}$

Q2. As $\frac{d}{dx}(\cos^{-1}x) = -\frac{d}{dx}(\sin^{-1}x)$

so not necessary

Q3. a) $\int \frac{1}{\underbrace{25}_{a^2}+x^2}\,dx = \frac{1}{5}\tan^{-1}\left(\frac{x}{5}\right)$

$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$

b) $\int \frac{1}{\sqrt{\underbrace{16}_{a^2}-x^2}}\,dx = \sin^{-1}\left(\frac{x}{4}\right)$

$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right)$

Competency skills

need a^2+x^2

Q4. a) $\int \frac{1}{125+5x^2}\,dx = \frac{1}{5}\int \frac{1}{25+x^2}\,dx$

$= \frac{1}{5} \times \frac{1}{5}\tan^{-1}\left(\frac{x}{5}\right) = \frac{1}{25}\tan^{-1}\left(\frac{x}{5}\right)$

b) $\int \frac{1}{\sqrt{40-2x^2}}\,dx = \int \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{20-x^2}}\,dx$

$= \frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{x}{\sqrt{20}}\right)$

Q5. a) $\int_0^1 \frac{1}{\sqrt{9-3x^2}}\,dx = \frac{1}{\sqrt{3}}\int_0^1 \frac{1}{\sqrt{3-x^2}}\,dx$

$= \frac{1}{\sqrt{3}}\left[\sin^{-1}\left(\frac{x}{\sqrt{3}}\right)\right]_0^1 = \frac{1}{\sqrt{3}}\left(\sin^{-1}\left(\frac{1}{\sqrt{3}}\right) - 0\right)$

$= \frac{1}{\sqrt{3}}\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

b) $\int_1^{\sqrt{3}} \frac{5}{3+x^2}\,dx = 5\int_1^{\sqrt{3}} \frac{1}{3+x^2}\,dx$

$= 5\left[\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)\right]_1^{\sqrt{3}} = \frac{5}{\sqrt{3}}\left(\tan^{-1}(1) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$

$= \frac{5}{\sqrt{3}}\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \frac{5}{\sqrt{3}} \times \frac{\pi}{12} = \frac{5\pi}{12\sqrt{3}}$

c) $\int_0^{\frac{1}{2}} \frac{x}{-2x^4-60}\,dx = -\frac{1}{2}\int_0^{\frac{1}{2}} \frac{x}{x^4+30}\,dx$

sub $u = x^2$ so $u' = 2x$ so $dx = \frac{1}{2x}du$

$= -\frac{1}{4}\int_0^{\frac{1}{4}} \frac{1}{u^2+30}\,dx = -\frac{1}{4}\left[\frac{1}{\sqrt{30}}\tan^{-1}\left(\frac{u}{\sqrt{30}}\right)\right]_0^{\frac{1}{4}}$

$= -\frac{1}{4}\left(\frac{1}{\sqrt{30}}\tan^{-1}\left(\frac{1}{4\sqrt{30}}\right) - 0\right) = -\frac{1}{4\sqrt{30}}\tan^{-1}\left(\frac{1}{4\sqrt{30}}\right)$

Q6. $\displaystyle\int_{-\frac{9}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{25-(2x+4)^2}}\,dx$

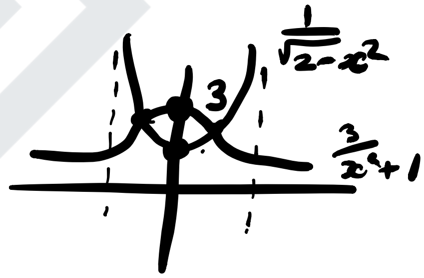
Sub $u = 2x+4$ $\quad u' = 2$ so $dx = \frac{1}{2}du$

$$= \int_{-5}^{5} \frac{1}{\sqrt{5^2-u^2}} \times \frac{1}{2}du = \frac{1}{2}\left[\sin^{-1}\left(\frac{u}{5}\right)\right]_{-5}^{5}$$

$$= \frac{1}{2}\left(\sin^{-1}(1) - \sin^{-1}(-1)\right) = \frac{1}{2}\left(\frac{\pi}{2} - -\frac{\pi}{2}\right)$$

$$= \frac{1}{2} \times \pi = \frac{\pi}{2}$$

Q7. $y = \dfrac{3}{x^2+1}$, $y = \dfrac{1}{\sqrt{2-x^2}}$



For intersections $\dfrac{3}{x^2+1} = \dfrac{1}{\sqrt{2-x^2}} \Rightarrow 3\sqrt{2-x^2} = x^2+1$

$\Rightarrow 9(2-x^2) = (x^2+1)^2 \Rightarrow 18 - 9x^2 = x^4 + 2x^2 + 1$

$\Rightarrow \quad x^4 + 11x^2 - 17 = 0 \Rightarrow x^2 = \dfrac{-11 \pm \sqrt{121+68}}{2}$

As $x^2 > 0$ $\quad x^2 = \dfrac{-11+\sqrt{189}}{2}$ so $x = \pm\sqrt{\dfrac{\sqrt{189}-11}{2}}$

so $\displaystyle\int_{-\sqrt{\frac{\sqrt{189}-11}{2}}}^{\sqrt{\frac{\sqrt{189}-11}{2}}} \frac{3}{x^2+1} - \frac{1}{\sqrt{2-x^2}}\,dx$

y as symmetric in x

$$= 2\int_0^{\sqrt{\frac{\sqrt{189}-11}{2}}} \frac{3}{x^2+1} - \frac{1}{\sqrt{2-x^2}}\,dx$$

$$= 2\left[3\tan^{-1}(x) - \sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\right]_0^{\sqrt{\frac{\sqrt{189}-11}{2}}}$$

$$= 2\left(3\tan^{-1}\left(\sqrt{\frac{\sqrt{189}-11}{2}}\right) - \sin^{-1}\left(\frac{\sqrt{\sqrt{189}-11}}{2}\right) - 0\right)$$

$$= 6\tan^{-1}\left(\sqrt{\frac{\sqrt{189}-11}{2}}\right) - 2\sin^{-1}\left(\frac{\sqrt{\sqrt{189}-11}}{2}\right)$$

Advanced skills

Q8. a) $\displaystyle\int_{-1}^{2} \frac{1}{\sqrt{4x+5-x^2}}\,dx = \int_{-1}^{2} \frac{1}{\sqrt{9-(x-2)^2}}\,dx$

complete square

sub $u = x-2$

$$= \int_{-3}^{0} \frac{1}{\sqrt{9-u^2}}\,du$$

$$= \left[\sin^{-1}\left(\frac{u}{3}\right)\right]_{-3}^{0} = \sin^{-1}(0) - \sin^{-1}(-1)$$

$$= 0 - -\frac{\pi}{2} = \frac{\pi}{2}$$

b) $$\int \frac{1}{20-6x+2x^2}\,dx = \frac{1}{2}\int \frac{1}{10-3x+x^2}\,dx$$

$$= \frac{1}{2}\int \frac{1}{7.75+(x-1.5)^2}\,dx = \frac{1}{2}\int \frac{1}{7.75+u^2}\,du$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{7.75}} \tan^{-1}\left(\frac{u}{\sqrt{7.75}}\right) = \frac{1}{2\sqrt{7.75}} \tan^{-1}\left(\frac{x-1.5}{\sqrt{7.75}}\right)$$

Q9. $\int \frac{1}{\sqrt{1-x^2}}\,dx$ let $x = \sin u$

so $dx = \cos u\,du$

$$= \int \frac{1}{\sqrt{1-\sin^2 u}} \times \cos u\,du = \int \frac{\cos u}{\cos u}\,du$$

$\cos^2 u + \sin^2 u = 1$

$$= \int 1\,du = u = \sin^{-1}(x)$$

$x = \sin u$

Q10. $\int \frac{1}{1+x^2}\,dx$ $x = \tan u \Rightarrow dx = \sec^2 u\,du$

$$= \int \frac{1}{1+\tan^2 u} \times \sec^2 u\,du = \int \frac{\sec^2 u}{\sec^2 u}\,du = \int 1\,du = u$$

$1+\tan^2 u = \sec^2 u$

$$= \tan^{-1}(x)$$

Q11. $$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx$$

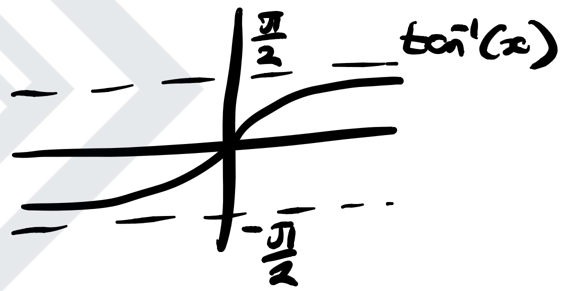
$$= \lim_{k \to -\infty} \left[\tan^{-1}(x)\right]_k^0 + \lim_{k \to \infty} \left[\tan^{-1}(x)\right]_0^k$$

$$= \lim_{k \to -\infty} \left(\tan^{-1}(0) - \tan^{-1}(k)\right) + \lim_{k \to \infty} \left(\tan^{-1}(k) - \tan^{-1}(0)\right)$$

$$= \lim_{k \to -\infty} -\tan^{-1}(k) + \lim_{k \to \infty} \tan^{-1}(k)$$

$$= - -\frac{\pi}{2} + \frac{\pi}{2}$$

$$= \pi$$



Q12. $\int \sqrt{1-x^2}\,dx$ let $x = \sin u$

so $dx = \cos u\,du$

$$\text{so } \int \sqrt{1-\sin^2 u}\,\cos u\,du = \int \cos^2 u\,du$$

$\cos^2 + \sin^2 = 1$

$\cos 2u = 2\cos^2 u - 1$

$$= \int \frac{1}{2} + \frac{1}{2}\cos 2u\,du = \frac{1}{2}u + \frac{1}{4}\sin 2u$$

$\sin 2u = 2\sin u \cos u$

$$= \frac{1}{2}u + \frac{1}{2}\sin u \underbrace{\cos u} = \frac{1}{2}\sin^{-1}x + \frac{1}{2}x\sqrt{1-x^2}$$

as $\sin u = x$

$\cos u = \sqrt{1-x^2}$ by $\sin^2 + \cos^2 = 1$